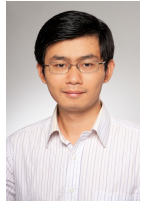


Curriculum Vitae

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EDUCATION

- | | |
|---|--|
| • Leibniz University Hannover
<i>PhD candidate in the Institute of Artificial Intelligence</i> | <i>Hannover, Germany
Mar. 2020 - Present</i> |
| • Technical University of Darmstadt
<i>Master of Science in Electrical Engineering and Information Technology</i> | <i>Darmstadt, Germany
Oct. 2015- Sep. 2019</i> |
| • Huazhong University of Science and Technology
<i>B.E. in Electronics and Information Engineering</i> | <i>Wuhan, China
Sep. 2011- July 2015</i> |

RESEARCH DIRECTION

- Efficient Deep Learning & Transformer Architectures
- Neural Architecture Search
- AutoML & Hyperparameter Optimization
- Time Series Forecasting

SKILLS

- Python, Machine Learning, Deep Learning
- Tabular & Structured Data Modeling
- Sequential Data Modeling
- Distributed Multi-GPU Training Pipelines
- GPU Programming and Kernel Optimization

OPEN-SOURCE PROJECTS

Efficient Language Models

Neural Attention Search Linear (NAtS-L)¹

<https://github.com/automl/NeuralAttentionSearchLinear>

Role: Developer

NAtS-L is a token-level hybrid attention framework designed for training efficient long-context large language models. Implemented in Triton and trained across multiple GPUs, the framework introduces a learnable gating mechanism (similar to Mixture-of-Experts routing process) that dynamically selects attention operations from a learned mixture of full and linear attention kernels based on the input feature map, enabling input-adaptive computation across layers. This architectural design directly aims at both pre-training and inference efficiency: the hybrid formulation reduces FLOPs during both pre-training and inference while maintaining model quality, achieving $5.4\times$ speedup in pre-filling and $2.3\times$ in decoding latency over full-attention baselines, with up to 34% improvement on long-context retrieval benchmarks.

Neural Attention Search (NAtS)²

<https://github.com/automl/NeuralAttentionSearch>

Role: Developer

NAtS is a framework for learning optimal sparse attention patterns during pre-training of large language models, implemented in Triton for efficient kernel designs. NAtS introduces a learned router (architecturally equivalent to MoE gating) that maps input representations to sparse attention operation types, enabling the model to adaptively allocate KV cache capacity per token and layer. By retaining only 15%–20% of KV cache entries while preserving 96% of full-attention performance, NAtS reduces memory consumption by $2.24\times$ and achieves $3\times$ latency speedup at inference.

Time Series Forecasting

¹Related Paper: <https://arxiv.org/abs/2602.03681>, accpeted at ICML 2026

²Related Paper: <https://openreview.net/forum?id=h3gaIBwemp>, accepted at NeurIPS 2025

Auto-PyTorch for Time Series Forecasting (APT-TS)³

<https://github.com/automl/Auto-PyTorch>

Role: Developer

AutoPyTorch-TS (2.5k GitHub stars) is a widely adopted open-source automated deep learning framework for time series forecasting. From data ingestion and feature engineering through architecture selection and hyperparameter optimization, it constructs end-to-end forecasting pipelines tailored to the input data distribution. The framework supports high-dimensional, heterogeneous multivariate datasets with minimal search overhead and a modular design that enables rapid iteration across forecasting horizons and data modalities.

DARTS-TS⁴

<https://github.com/automl/OneShotForecastingNAS>

Role: Developer

DARTS-TS is an open-source neural architecture search framework for time series forecasting that automatically discovers task-specific deep learning architectures across diverse operation families, including recurrent, convolutional, and attention-based modules. Designed for production constraints, it delivers inference latency on par with hand-crafted baselines while removing manual architecture engineering. Directly applicable to multi-horizon demand forecasting in different task settings that involve heterogeneous, multivariate time series, enabling scalable and maintainable forecasting solutions without manual model tuning.

Hyperparameter Optimization

SMAC3⁵

<https://github.com/automl/SMAC3>

Role: Contributor

SMAC3 (1.2k GitHub stars) is a widely used open-source Bayesian Optimization library for hyperparameter optimization and AutoML. Its highly modular architecture allows users to freely assemble components into custom optimization pipelines. The Bayesian modeling approach is directly transferable to inventory optimization problems such as replenishment policy search and procurement parameter tuning under uncertainty.

SELECTED PUBLICATIONS

- **Deng, D., Winje, A., Fehring, L., Lindauer, M. (2026).** *Neural Attention Search Linear: Towards Adaptive Token-Level Hybrid Attention Models.* in *Forty-Third International Conference on Machine Learning (ICML 2026)*
<https://arxiv.org/abs/2602.03681>
– Keyword: Efficient LLM, Linear Attention, Hybrid Attention
- **Deng, D. and Lindauer, M. (2025).** *Neural Attention Search.* in *39th Conference on Neural Information Processing Systems (NeurIPS 2025)*. <https://openreview.net/forum?id=h3gaIBwemp>
– Keyword: Efficient LLM, Sparse Attention, KV Cache Reduction, Parameter-Efficient Fine-Tuning
- **Deng, D. and Lindauer, M. (2025).** *Optimizing Time Series Forecasting Architectures: A Hierarchical Neural Architecture Search Approach.* *Transactions on Machine Learning Research.*
<https://openreview.net/forum?id=Ym2wqojm4e>
– Keyword: Multi-Variant Time Series Forecasting, Neural Architecture Search, Automated Machine Learning
- **Deng, D., Karl, F., Hutter, F., Bischl, B., and Lindauer, M. (2022).** *Efficient Automated Deep Learning for Time Series Forecasting.* in *Proceedings of the European Conference on Machine Learning (ECML)*.
https://dl.acm.org/doi/10.1007/978-3-031-26409-2_40
– Keyword: Time Series Forecasting, Neural Architecture Search, Automated Machine Learning
- **Tornede, A., Deng, D., Eimer, T., Giovanelli, J., Mohan, A., Ruhkopf T., Segel S., Theodorakopoulos, D., Tornede, T., Wachsmuth, H., Lindauer, M. (2023).** *AutoML in the Age of Large Language Models: Current Challenges, Future Opportunities and Risks.* *Transactions on Machine Learning Research.*
<https://openreview.net/forum?id=cAthubStyG>
- **Lindauer, M., Eggensperger, K., Feurer, M., Biedenkapp, A., Deng, D., Benjamins, C., Sass, R., and Hutter, F. (2022).** *SMAC3: A Versatile Bayesian Optimization Package for Hyperparameter Optimization.* *Journal of Machine Learning Research*, 2022(23). <https://www.jmlr.org/papers/volume23/21-0888/21-0888.pdf>

AWARDS

- 2020: 3rd place(*) at the official leaderboard and 1st place at the warmstart friendly leaderboard at the BBO-Challenge at NeurIPS'20 (* after fixing a minor bug)

³Related Paper: https://dl.acm.org/doi/10.1007/978-3-031-26409-2_40

⁴Related Paper: <https://openreview.net/forum?id=Ym2wqojm4e>

⁵Related Paper: <https://www.jmlr.org/papers/volume23/21-0888/21-0888.pdf>